

A finite pencil criterion for boundary-norm graph obstructions

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Abstract

Boundary arguments for polynomial graph sections often produce arbitrarily high-degree homogeneous forms, apparently requiring a new obstruction in every degree. We show that, under a precise dicritical-cover package, the entire all-degree problem is equivalent to one finite calculation in a two-dimensional polynomial pencil. The key field-norm lemma is independent of the degree of the covering dicritical: if both s and ℓ pull back to polynomials on an affine-line cover and s^κ/ℓ is polynomial, then the boundary constant term is a monomial cs^q with $q \leq \kappa \deg_\ell F$. Factoring a homogeneous form into linear factors then reduces every graph degree to whether $\mathbb{C}A + \mathbb{C}B$ contains one of $1, s, \dots, s^\kappa$, up to scalar. We formulate a conditional boundary-norm transfer theorem, recover the graph obstruction for the six-sheeted graded Keller seed of Kistner and Shaska, and prove that the pencil hypothesis is sharp for this method. The theorem does not establish the geometric boundary package for arbitrary maps and does not decide the two-dimensional Jacobian conjecture.

1. Motivation and result

Let $k = \mathbb{C}$. Consider a polynomial graph equation

$$W + H(U, V) = 0, \quad H(0, 0) = 0.$$

In the six-sheeted graded Keller example of Kistner and Shaska [1], a boundary chart for a hypothetical affine-plane inverse-image component has residual target functions

$$U = \frac{s^\kappa A(s)}{\ell}, \quad V = \frac{s^\kappa B(s)}{\ell}, \quad (1.1)$$

with $\kappa = 2$. The accepted case-specific proof factors the special fibre of the graph closure and applies field norms on covering affine-line dicriticals. This article isolates the algebra that actually drives the final contradiction.

The main conceptual result is that no enumeration over $\deg H$ is needed. Once the geometric boundary package has been proved for a seed, the all-degree obstruction is equivalent to the finite condition

$$(kA + kB) \cap \{cs^q : c \in k^\times, 0 \leq q \leq \kappa\} = \emptyset. \quad (1.2)$$

This is an extension of the fixed-seed proof, not a solution of a Jacobian case. In particular, the hypotheses below are not asserted for every graded Keller map.

2. A cover-degree-free norm lemma

Let s, ℓ be indeterminates and let $\kappa \geq 0$. Let

$$F(s, \ell) \in k[s, \ell]$$

be irreducible and monic of positive ℓ -degree n . Put

$$f(s) = F(s, 0) \neq 0, \quad K = \text{Frac}(k[s, \ell]/(F)).$$

Assume there is a finite extension

$$K \hookrightarrow L = k(t)$$

of degree e such that the images $s(t)$ and $\ell(t)$ are polynomials, $s(t)$ is nonconstant, and

$$h(t) = \frac{s(t)^\kappa}{\ell(t)} \in k[t]. \quad (2.1)$$

Lemma 2.1 (norm-support bound)

Under these assumptions,

$$\boxed{f(s) = cs^q, \quad c \in k^\times, \quad 0 \leq q \leq \kappa n.} \quad (2.2)$$

Proof Because $s(t)$ is a nonconstant polynomial, t is integral over $k[s]$: divide $s(t) - s$ by the nonzero constant leading coefficient of $s(t)$. Thus $k[t]$ is finite integral over $k[s]$, and the polynomial $h(t)$ is integral over $k[s]$.

Its field norm $N_{L/k(s)}(h)$ is integral over $k[s]$ and lies in $k(s)$. Since the PID $k[s]$ is integrally closed,

$$N_{L/k(s)}(h) \in k[s]. \quad (2.3)$$

Monicity and Gauss's lemma give $[K : k(s)] = n$, while

$$N_{K/k(s)}(\ell) = (-1)^n f(s).$$

Since $h = s^\kappa/\ell$ belongs to K , the tower law gives

$$N_{L/k(s)}(h) = \left(N_{K/k(s)}(h)\right)^e = \pm \frac{s^{\kappa en}}{f(s)^e} \in k[s]. \quad (2.4)$$

Hence $f(s)^e$ divides a power of s in the UFD $k[s]$. Every irreducible factor of f is therefore associated to s , proving $f = cs^q$. Comparing exponents in (2.4) gives $eq \leq \kappa en$, and the cover degree e cancels. $\square$

The cancellation of e is useful: the boundary curve itself need not have affine-line normalization. It suffices to have a finite dominant affine-line dicritical above it.

3. Every homogeneous degree reduces to one pencil

For coprime nonzero $A, B \in k[s]$, define

$$\mathcal{M}_\kappa = \{cs^q : c \in k^\times, 0 \leq q \leq \kappa\}.$$

Lemma 3.1 (finite pencil equivalence)

The following are equivalent.

1. $(kA + kB) \cap \mathcal{M}_\kappa = \emptyset$.
2. For every $m \geq 1$ and every nonzero homogeneous $H_m \in k[X, Y]$ of degree m ,

$$H_m(A(s), B(s)) \notin \mathcal{M}_\kappa. \quad (3.1)$$

Proof The second statement implies the first by taking $m = 1$. Conversely, factor over the algebraically closed field:

$$H_m(X, Y) = c \prod_{j=1}^m (a_j X + b_j Y).$$

Suppose $H_m(A, B) = c' s^q$ with $q \leq \kappa$. No factor $a_j A + b_j B$ is zero, since the product is nonzero. Unique factorization in $k[s]$ forces every factor to be $c_j s^{q_j}$, with $q_j \geq 0$ and $\sum_j q_j = q$. Thus at least one pencil element belongs to $\mathcal{M}_\kappa$, contradicting statement 1. $\square$

Corollary 3.2 (finite algorithm)

Condition 1 is decided by the finite family of linear systems

$$uA(s) + vB(s) = cs^q, \quad q = 0, \dots, \kappa. \quad (3.2)$$

No graph-degree cutoff is involved.

4. The geometric transfer package

The next theorem separates geometry from algebra. Let $H \in k[U, V] \setminus \{0\}$ satisfy $H(0, 0) = 0$, put $m = \deg H \geq 1$, and write H_m for its top homogeneous part. Suppose that, under a hypothetical identification of a selected graph inverse-image component with $\mathbb{A}^2$, one has proved the following data.

Boundary-norm hypotheses $\text{BN}(\kappa; A, B)$

1. There is an integral saturated partial boundary chart with parameters (s, ε, ℓ) . If $m = \deg H$, its scheme-theoretic special fibre at $\varepsilon = 0$ is the monic polynomial

$$C_H(s, \ell) = \prod_i F_i(s, \ell)^{r_i}, \quad \sum_i r_i n_i = m + 1, \quad (4.1)$$

where the F_i are distinct irreducible monic polynomials and $n_i = \deg_\ell F_i \geq 1$.

2. Its constant term is exactly

$$C_H(s, 0) = H_m(s^\kappa A(s), s^\kappa B(s)) \neq 0. \quad (4.2)$$

3. Every reduced factor is a genuine horizontal boundary prime and has regular nonconstant target residues

$$U = \frac{s^\kappa A(s)}{\ell}, \quad V = \frac{s^\kappa B(s)}{\ell}. \quad (4.3)$$

4. Every factor is dominated by a finite affine-line dicritical $\mathbb{A}_t^1 \rightarrow V(F_i)$ on which $s(t)$ and $\ell(t)$ are polynomials.

5. $\gcd(A, B) = 1$. Hence Bézout's identity and (4.3) make

$$h = \frac{s^\kappa}{\ell}$$

polynomial on each affine-line dicritical.

These hypotheses are deliberately explicit. Saturation, boundary exhaustion, regular residues and dicritical domination are proof obligations, not automatic consequences of writing down a chart.

5. Boundary-norm transfer

Theorem 5.1

If $\text{BN}(\kappa; A, B)$ holds and

$$(kA + kB) \cap \mathcal{M}_\kappa = \emptyset, \quad (5.1)$$

then the selected graph component cannot be isomorphic to $\mathbb{A}^2$.

Proof Put $f_i(s) = F_i(s, 0)$. Hypotheses 3–5 allow Lemma 2.1 to be applied to every factor in (4.1). Thus

$$f_i(s) = c_i s^{q_i}, \quad 0 \leq q_i \leq \kappa n_i. \quad (5.2)$$

Taking constant terms in (4.1) and using (4.2),

$$s^{\kappa m} H_m(A, B) = cs^Q, \quad Q = \sum_i r_i q_i \leq \kappa \sum_i r_i n_i = \kappa(m+1). \quad (5.3)$$

The left side is nonzero and divisible by $s^{\kappa m}$, so $Q \geq \kappa m$. Consequently

$$H_m(A, B) = cs^q, \quad 0 \leq q = Q - \kappa m \leq \kappa.$$

This contradicts Lemma 3.1 and (5.1). $\square$

The theorem is seed-independent but conditional. Its value is that after the geometric package has been established, the unbounded family of graph degrees collapses to the finite pencil test (3.2).

6. Six-sheeted application

For the Kistner–Shaska six-sheeted seed [1, Theorem 3.19], the companion boundary-chart calculation gives $\kappa = 2$ and

$$A(s) = \alpha + (1 - \alpha)s, \quad (6.1)$$

$$B(s) = s \left(\frac{8}{3}\alpha^2 + \frac{9}{2}\alpha(1 - \alpha)s + \frac{9}{5}(1 - \alpha)^2s^2 \right), \quad \alpha^2 + 3\alpha + 6 = 0. \quad (6.2)$$

The full chart proof establishes $\text{BN}(2; A, B)$ for every nonzero graph polynomial H with $H(0, 0) = 0$; it is included byte-for-byte as the accepted companion six-sheet graph manuscript in the same evidence package. Its canonical Markdown SHA-256 is `c7fc0e7ed904504e1b7a7ec01e4a22c867b9d9b2bfe9978bc23619c794240010`. The required algebra is short:

$$\text{Res}_s(A, B) = 2\alpha \neq 0,$$

so A and B are coprime. If $uA + vB$ were a monomial of degree at most two, its cubic coefficient would force $v = 0$, because $(9/5)(1 - \alpha)^2 \neq 0$. The remaining polynomial uA has both a nonzero constant and a nonzero linear coefficient unless $u = 0$. Therefore

$$(kA + kB) \cap \{c, cs, cs^2 : c \in k^\times\} = \emptyset. \quad (6.3)$$

Theorem 5.1 recovers the all-degree graph obstruction. An exact SymPy checker over $\mathbb{Q}(\sqrt{-15})$ verifies the field relation, resultant, gcd and all three pencil systems under both normal and optimized Python.

This application is distinct from Proposition 3.21 of [1]. That proposition shows that the three-variable graded family does not directly become a two-variable polynomial pair after the substitution $T = \Lambda w$, because a top term has a negative Λ -exponent. The present theorem concerns polynomial graph sections, finite boundary normalizations and field norms.

7. Sharp limitation of the method

The pencil condition cannot be weakened within the norm/top-form argument. Take

$$A = 1, \quad B = s, \quad \kappa \geq 1.$$

Then A and B are coprime, but the pencil contains both 1 and s . The degree-one homogeneous forms X and Y already produce forbidden monomials. Thus condition (5.1) is necessary and sufficient for the all-degree conclusion of Lemma 3.1.

This is a counterexample to weakening the algebraic method, not a Keller-map counterexample and not a counterexample to Theorem 5.1. A seed whose pencil meets $\mathcal{M}_\kappa$ may still be obstructed by other geometry.

8. Literature position and limitations

Orevkov’s boundary-chain and multiplicity arguments [2] and Borisov’s Stein factorization theorem [3] provide strong global restrictions on hypothetical plane Keller maps. In particular, Borisov already proves much of the one-point boundary and normalization geometry that appeared broadly in the predecessor P4 manuscript. We do not claim those global facts as new.

The contribution isolated here is narrower: the cover-degree-free exponent bound, the finite-pencil equivalence, and their conditional transfer through a precise graph-boundary package. Current exact-phrase, citation-chain and arXiv searches found no equivalent graph-section theorem, but this is not a proof of priority.

The limitations are substantive:

- $\text{BN}(\kappa; A, B)$ must be proved for each application;
- the theorem does not cover arbitrary non-graph surfaces or arbitrary target coordinate changes;
- the six-sheet application depends on the companion chart-to-dicritical proof, not merely on the displayed pencil calculation;
- the Kistner–Shaska seed paper is a current unrefereed arXiv v1; and
- no conclusion about the two-dimensional Jacobian conjecture follows.

References

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- [2] S. Yu. Orevkov, “On three-sheeted polynomial mappings of $\mathbb{C}^2$,” *Mathematics of the USSR-Izvestiya* **29** (1987), 409–430. <https://doi.org/10.1070/IM1987v029n03ABEH000984>.
- [3] A. Borisov, “On the Stein factorization of resolutions of two-dimensional Keller maps,” *Beiträge zur Algebra und Geometrie* **56** (2015), 299–312. <https://doi.org/10.1007/s13366-014-0208-4>.

Declarations

Data and code availability. The companion geometric proof, exact pencil checker, receipts and frozen manifest accompany the candidate article.

Author responsibility. The submitting author is responsible for every mathematical and historical claim.

AI-use disclosure. AI systems contributed to source discovery, theorem extraction, proof exploration, exact checking, review and drafting. Internal AI review is not external peer review.